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	Three projects within the ALS NSCORT address the feasibility of producing human-edible biomass on Mars or the moon: One project uses lignin-degrading fungi to produce exotic mushrooms; another project examines growth of fresh-water fish on substrates derived from crops and food scraps (Fish in Space: High Quality Food Production Coupled with Minimization of Equivalent System Mass; project discussed in the "Solid Waste"/ "Resource Recovery " section of the ALS NSCORT report) and a third is proving that it is energetically feasible to grow crops with electric lighting in space. The rationale for using heterotrophic fish and fungi in space is both to reduce the solid-waste burden and improve the nutrition and interest of crews in a diet that will be mainly vegetarian once life-support loops are largely closed. The ESM crossover point when fish and mushroom production would begin to compete with human crews for resources such as oxygen will determine the upper limit for using them as alternative food sources or as waste processors.		

Task Description:	During the past year, the Center has broadly embraced the concept that resource recovery from bioprocessing must identify potential hazards to crews in a closed life-support system as well as be kinetically and energetically efficient. Whenever processor organisms enzymatically digest organic substrates, such as in a microbial bioreactor, it is legitimate to ask whether the effluents and residues contain toxic substances derived either from breakdown of the waste biomass itself, or as a by-product of microbial metabolism. Alternative food organisms feeding on substrates containing potential toxins need to be analyzed, as do those substrates, before and after bio-processing, to determine the type and degree of processing needed to mitigate risk to crews in a closed system. Similar questions are being asked regarding the use of plants for waste processing. Emphasis has shifted in the NSCORT toward use of non-human wastes as substrates that make bioreactor effluents and residues potentially safer for subsequent feeding to fish and fungi that, in turn, could be consumed by human crews. The prospect of using white-rot fungi to pre-digest crop residue prior to feeding it to Tilapia is under investigation, either after composting and/or with an intermediate microbial bioreactor step involved. Focus is on meeting the nutritional requirements of fish as well as that of the bioreactor microorganisms they will feed on. In order for fish to obtain essential amino acids from N-deficient ligno-cellulose, bacterial biomass has been identified as a needed component of the bioreactor residue, and nitrogenous and sulfurous volatile effluents from other bioprocesses are to be captured and redirected to enhance bacterial, fungal, and fish nutrition and health. Both alternative food research projects are entering additional collaborations within the NSCORT to better enable fish and fungi to fill dual food and waste-processor roles for an energy-efficient ALS. The third biomass project is tackling the energy-burden issue of lighting crops with electric sources: One approach is to alter the architecture of crop lighting so that all photosynthetic surfaces within crop stands are equally targeted at all times; a second approach is to use relatively low power so that LED emitters can be positioned close to plant tissues without sacrificing irradiance incident upon those tissues or causing thermal stress; still another approach involves remotely sensing the position of growing plant tissues and selectively switching LED light engines on or off automatically to avoid lighting empty space or non-productive plant tissues; yet another approach uses only the narrow-spectrum radiation of selected LEDs that matches the most efficient absorption of light by crop plants. These approaches are being combined to develop an optimum lighting system with promise to significantly lower the ESM for electric lighting of crops in an ALS. Cary A. Mitchell Crop/Biomass Focus Area Lead For Bio Mass Production: 1 - Develop a reliable crop-lighting system for space that avoids the main disadvantages of traditional plant-growth lamps and reduces the ESM for crop production enough that NASA can consider putting plants back into near-term exploration missions. 2 - Avoid tissue damage by adopting solid-state, light-emitting-diode (LED) arrays that do not have hot light-emitting surfaces and which can be placed very close to plant tissues while maintaining PAR irradiance. 3 - Reduce the thermal burden of plant-growth lighting by operating the PAR sources at low power. 4 - Achieve uniform light distribution and maximize PAR absorption within crop canopies by arraying LEDs within or above crop stands so that all photosynthetic tissues are uniformly irradiated. 5 - Improve the quantum efficiency of absorbed light by using only wavelengths coinciding with maximum light absorption by photosynthetic pigments. 6 - Selectively target photons to be incident only upon photosynthetic tissue. For Lignin-Degrading Fungi: 1 - The objectives of our effort are to optimize biodegradation of inedible crop residues and production of mushrooms, to evaluate and select most efficient species and strains of edible white rot fungi, determine most efficient cropping patterns, nutrient requirements, amount of light, temperature, CO2, O2 and humidity required for optimal fungal colonization and fruiting. Both projects potentially extend the harvest index of crop production by converting lignocellulose to substrates for growth of intermediate organisms that can be eaten by humans. Fish, of course, could provide balanced protein and calcium to supplement a mostly vegetarian ALS diet, and mushrooms could psychologically augment crew diets through texture, flavor, some nutrition, and interest. The main benefit of white-rot fungi, however, might be to pre-treat and “soften” vegetation that fish have a hard time digesting. If tilapia find fungal-treated crop residue to be palatable, a positive synergism could occur between the two bridging projects and the crop-production project of the NSCORT. Since both “processor organisms” also are heterotrophic, the systems group, in collaboration with the fish, fungal, and crop labs, are calculating how much fish and fungal biomass production can be accommodated in an ALS before the benefits to humans are superceded by competition for resources such as oxygen.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Development of solid-state, intracanopy and close-canopy crop-lighting systems will solve the light-distribution problem that has long limited the photosynthetic productivity of overhead-lighted crop stands, and will permit the use of low-power light sources. These advances will enable year-round, warehouse-based, controlled-environment agriculture (CEA) production of salad greens, herbs, and certain vegetables to be profitable in local markets, especially if coupled with sources of cheap, bio-derived energy, and competitive with produced trucked from distant production sites. 2. Development of new water and air-treatment technologies for environmental protection 3. Will impact air-treatment technology and air quality control strategies used in residential and commercial buildings and in commercial aircraft. 4. Further development of fundamental photochemical reactor theory as it relates to systems used for disinfection of water and air. 5. Further development of biological methods for converting organic wastes into fuels, chemicals, and other useful products. 6. Further development of nano-scale detection of pathogens in packaged foods

7. Development of novel food products and technologies for vegetarian and health-food markets

8. Developing methods for maximum crop yield at minimum cost in controlled-environment agriculture (CEA)

9. Developing genetic engineering of CEA crops for superior performance

10. Development of computer models for control of complex processes

For Bio Mass Production:
LED Lighting Project: Trials 1-5. Since delivery of the first reconfigurable LED lighting array from Orbitec in mid August, 2004, we have run four hardware tests in the intracanopy (IC) configuration using the dry-bean crop cowpea (Vigna unguiculata (L.) Walp. IT87D-941-1). Tests ran for approximately 30 days each, and incremental improvements in crop response were made for each new trial as optimization of input parameters occurred. Plant spacing, plant density, electrical power level, red:blue ratio, reflective curtains, and thigmostimulation were parameters that were varied to evoke marked increases in biomass accumulation with each subsequent trial. As the crop stand grew in height during trials 1- 4, an increasing number of light engines were switched on along LED strips from the bottom up. In trials 3 and 4, a reflective false ceiling was installed in the growth compartment and raised incrementally to keep pace with increases in crop height and the number of engines energized. Daily power usage, therefore, increased as the experiment progressed. Trial 5 was performed with the lights reconfigured into an overhead, planar array. The growth conditions for trial 5 were identical to those of trial 4, with the exception of the lighting position and the daily power usage. The average daily power level for trial 4 was calculated, and this was used as the daily power level for trial 5, which corresponded to 55.2% of maximum power or a constant PAR of ~190µmols/m2/s at 2.5 cm below the lights. For all trials, photographic data of plant-growth characteristics, fresh and dry weights of stems, roots, leaves, leaf areas, and stem internode lengths were collected upon harvest. From the photographs it is clear that, in the overhead configuration (trial 5), lower leaf senescence occurred, which does not occur with the lights in an intracanopy configuration. We believe that this leaf drop is due to the low light levels occurring within a closed canopy, as mutual shading occurs from overhead leaves. When PAR levels were measured at the bottom of a mature cowpea canopy in trial 5, light was virtually undetectable (PAR = 2µmols/m2/s). Trial 5, under the same environmental conditions as trial 4, showed reduced productivity, 10.9 gDW/m2/day vs. 13.0 g/m2/day for trial 4. These data for trial five include the dry weight of leaves dropped from the canopy. When these additional leaves are removed (9.1g), indicating edible plant matter (for cowpea), the differences between the two trials are even more marked (9.7g/m2/day vs. 13.0g/m2/day). These data indicate that intracanopy lighting successfully applies light only to the plants, and uses more of the plant's photosynthetic capacity. Maintaining lower leaves in the stands grown with IC lighting indicates that those resources invested on the crop are not wasted, as in the overhead lighting configuration that leads to lower-leaf shading and leaf drop.

Trial 6 is underway with the lighting array in an overhead configuration. This trial differs from trial five by varying the light level over the duration of the trial. Although all light engines will be energized, the light level will increase as the plants grow in the same manner as in trial 4. A second reconfigurable lighting array, modified to allow half of the lightsicles to nestle under the others to provide more flexibility in configurations and closer spacing if necessary, will arrive in early October of 2005. Hardware delivery was delayed to allow Orbitec to correct an anomaly found in the first array. Both arrays will therefore be fully functional, with light outputs from a light engine independent of the number of other engines energized. Side-by-side experiments will be conducted comparing simultaneous overhead and intracanopy lighting. These experiments will rigorously test our hypothesis that intracanopy lighting increases crop yield and reduces ESM compared to an overhead lighting configuration for planophile crops.

LED Lighting Project: HELIAC. In January, 2005, Orbitec was awarded a Phase 1 SBIR for the development of "High Efficiency Lighting with Integrated Adaptive Control" or HELIAC, with the Purdue NSCORT as a sub-contractor. A hardware prototype-testing unit was developed at Orbitec in consultation with the crops focus area, to initially apply manual control, and later the option for set programs. The programs include patterns of energizing green LEDs mounted on individual light engines, with reflectance from plants in the growth compartment automatically detected by photodiodes on the unlit engines. The threshold of photodiode sensing can be set to different levels, as well as the power applied to green LEDs that flash. Also, proximity of the detector to the plant is adjustable. When reflectance above threshold is detected by a pair of photodiodes on a light engine, the LEDs on that engine will be energized upon completion of the program, indicating that a reflective object (such as a leaf) was detected in front of that particular location. The prototype was assembled on the framework of a light stick that can be used in either a horizontal or vertical position in the same way as our reconfigurable array elements. One difference for the array test elements is that only green LEDs and photodiodes were populated on the light engines, as opposed to operational engines that have these components as well as red and blue LEDs.

Upon Orbitec's completion of the prototype, it was delivered to Purdue University for plant-based testing. Over several weeks we performed tests with a variety of ALS candidate crop species, including cowpea, soybean, wheat, peanut, sweetpotato, and strawberry. Tests were performed with the test strip suspended vertically in front of a test plant or suspended horizontally overhead at different distances, different thresholds, different light levels, and using three different programmed flashing light patterns. In addition, the HELIAC device was installed in our intracanopy lighting array in place of one lightsicle during the exponential growth phase of a cowpea stand. Repeated testing occurred at different stages of crop development. The data indicated how closely the observed pattern of detection mimicked human observation. Photographic data were also recorded. The final SBIR report submitted to NASA by Orbitec in July, 2005 indicated the results of those trials. Application for a Phase 2 SBIR was subsequently submitted that, will, if funded, allow integration of the detection apparatus into the reconfigurable lighting array to allow IC automation of light-engine switching as well as enable close-canopy (CC) studies selectively switching on light engines in a plane spaced closely above rosette or erectophile crops. CC development requires the automated switching capacity of HELIAC in addition to the independent light-engine switching proposed in the phase 2 to keep pace with plant growth. IC lighting will also improve dramatically in efficiency if select light engines can be turned off, and if automation allows precise switching control.

Secondary Projects. Secondary Biomass/Crop Production projects include collaborations with other NSCORT research groups, cultivar selection trials for ALS candidate crops, and manipulation of growth habits that could lead to a reduction in ESM for various candidate crops.

Task Progress:

Sweetpotato Cultivation Studies: Single vine training. After a series of greenhouse experiments examining effects of root restriction (container volume) and canopy size (controlled by vine pruning), we determined that sweetpotato (Ipomea batatas (Lam.) L. TU-82-155) storage root production was strongly inhibited by shoot pruning, stimulated by root restriction, and influenced by season. Nutritive aspects of the roots, however, were not consistently changed by any of these treatments. We then asked if vine training could be used in lieu of pruning to reduce the volume of space occupied by sweetpotato shoots. By maintaining each plant as a single vine, we were able to coil the vines around vertical cylindrical or conical frames to conserve the area required for vine growth. This reduces the light requirement (per m2 and per m3) greatly, without significantly influencing yield. Coupled with earlier studies showing that planting two vines in one pot significantly increased the per-pot yield, we are now planning to further optimize space utilization and light conservation by coiling two vines around a frame in opposite directions, essentially interweaving them. This vertical vine-training system lends itself well to intracanopy lighting strategies for this crop.

Strawberry cultivar selection and pollination: Temperature studies 1.Two plants each of four different day-neutral strawberry (Fragaria sp.L.) cultivars (‘Tristar’, ‘Tribute’, ‘Seascape’, and ‘Fern’) were grown in three reach-in growth chambers under the following temperature combinations: 15°C days/10°C nights, 18°C days/13°C nights, and 21°C days/16°C nights, as well as the greenhouse, which was set at 26.7°C days and 23.9°C nights. This trial ran from 1/26/05 to 5/12/05 and over that time, in general, the highest productivity obtained was from plants in the greenhouse. Plants in the chambers were exposed to approximately 330 µmol/m2/s of light from fluorescent and incandescent lamps, while plants grown in the greenhouse received generally high, but variable, natural light.

Strawberry cultivar selection and pollination: Temperature studies 2. New strawberry plants were ordered and started (6/2/05) under four different growth conditions. Five cultivars (‘Tribute’, ‘Tristar’, ‘Seascape’, ‘Fern’, ‘Cavendish’) were obtained by donation from the Indiana Berry and Plant Company. Plants were planted either in the greenhouse or in one of three reach-in Conviron growth chambers under three different day/night temperature (in °C) regimes (14 hr days, 18/10°, 21/13°, 24/16° day/night temperature, ramping RH 70% day/80% night, with incandescent and fluorescent lighting at 420 µmol/m2/s). Other plants were planted in the greenhouse. Growth chamber plants are maintained separately to prevent insect contamination and the need for pesticide application, so that fruit can be tested for flavor.

Strawberry cultivar selection and pollination: Pollination. Our previous strawberry studies involved no hand pollination and minimal flower shaking only occasionally. Although we had some decent fruit formation, we have decided to test pollination methods and do a crew time vs. yield trade-off study. All plants in the growth chambers are being hand pollinated, with pollen collected and pollination performed each Monday, Wednesday, and Friday. In the greenhouse, we have divided the plants into three groups of ten (two plants of each cultivar per group). One group remains untouched except for fruit harvesting and plant maintenance (removal of runners and dead leaves). A second group is pollinated with a brief (10 sec or less) touch of a vibrating pollination wand to each flower pedicel. The third group is hand pollinated with pollen collected from the growth-chamber plants. Pollen is not collected from the greenhouse due to pest contamination concerns. We have a low level of insect pests in the greenhouse, and plants there are periodically treated with pesticides at the demands of other facility users. Thus far, there are no pests in the chambers, which means that fruit from the chambers are available for consumption, and other flavor analyses.

pH control system. In collaboration with Dr. George Chiu, Moeed Mukhtar, a graduate student in Dr. Chiu’s lab, has developed a novel pH control system that is reaction invariant. This system has been fully tested without plants, and plant-based-testing is commencing in a recently refurbished EGC walk-in growth chamber.

Outreach. A tour was given to a group of Indiana artists interested in agriculture in Oct., 2004. This was part of a program organized by Agriculture Communications, and the artists were fascinated with the work being done in the ALS NSCORT. Additionally, a group of students and observers involved in the Biomass Production in Education program visited and toured facilities as part of the BPES symposium in Dec., 2004.

The crops focus-area group also spoke with a group of chemistry education software designers and artists regarding requirements for a self-sustaining life-support system in June, 2005. This consultation pertained to the design of a game to teach chemistry to high school students. The virtual base will be located on earth underground and will mimic a life-support system for the moon or mars.

The ALS NSCORT worked with the Purdue Agriculture Communications Exhibit Shop to develop an exhibit “Mission to Mars” that was located at the Indiana Sate Fair in Aug, 2005. In addition to contributing text, pictures, and layout advice for the exhibit, through a series of meetings and frequent communications, the crops focus area also developed, tested, and delivered an inexpensive hydroponic system as an integral part of the display. The hydroponic system is designed to be built by students, and in the exhibit it was planted with ‘Microtina’ tomatoes and cuttings of basil. This system was transparent but contained, so that all working components could be viewed easily but not disturbed by the public.

For Lignin-Degrading Fungi:

Pairing crop residue to enhance fungal growth and biodegradation of recalcitrant lignocellulosic waste

Previously, it was shown that pairing sweet potato or basil with wheat or rice straw improved mycelial growth and degradation of both sweet potato and basil. The goal of this study was to optimize white rot fungal growth and biodegradation of sweet potato and basil crop residues through pairing with wheat or rice straw. Our objective was to determine the best ratios in the crop pairing that would support optimal edible fungal growth and maximize degradation of the difficult crop residue in the mixture. Edible fungal strains of two species - P. ostreatus (‘Pohu’, ‘Grey Dover’) and P. eryngii, maintained as mycelial spawns on culture media in Petri dishes, at room temperature (23±2°C) in the laboratory were used. Rice straw, wheat straw, sweet potato, and basil were milled to pass through a 2mm sieve. The milled residues were weighed, placed in 750 ml food containers and thoroughly mixed by stirring with a glass rod. Sweet potato residue was combined with wheat or rice straw at various ratios (0:100, 20:80, 40:60, 60:40, 80:20, 100:0 [w/w]), and basil with rice straw at similar ratios to obtain a final weight of 80g per container. The substrate combinations were sufficiently moistened by adding 125ml distilled deionized water to each container. All cultures were autoclaved at 121°C and 1.1kgm-2 for 30 min. The sweet potato/rice and sweet potato/wheat residues were seeded with 3x3mm agar blocks of pure cultures of Pleurotus eryngii or P. ostreatus (‘Grey Dover’) while the basil/rice was inoculated with P. ostreatus (‘Pohu’). Inoculated cultures were capped and incubated in the dark for up to 21 days.

In the sweet potato/rice straw or sweet potato/wheat straw combination, a 20:80 ratio was most favorable for growth and degradation of sweet potato by P. eryngii . However, when P. ostreatus strain ‘Grey Dover’ was used, up to 60% sweet

potato could be incorporated into the mixture with rice straw or 40% with wheat straw and still obtain substantial degradation. ‘Grey Dover’ is more efficient and more tolerant of sweet potato than other species of edible white-rot fungi (EWRF) we have tested. Pairing sweet potato or basil with wheat or rice straw helped to diminish the impact of the inhibitory compounds present in the residues to allow fungal growth and colonization of the residues. Similarly, mycelial growth and colonization of P. ostreatus ‘Pohu’ was better in a 20:80 or 40:60 combination of basil with rice straw at 14 days after inoculation. Mycelial growth and colonization of recalcitrant crop residue by EWRF can be enhanced by combining the residues with wheat or rice straw at defined percentages of incorporation.

Both rice straw and wheat straw residues had promotive effects on fungal growth while alleviating the inhibitory effect of sweet potato or basil in the different substrate mixtures.

Optimizing edible fungal growth and degradation of inedible crop residues using various cropping methods

The objective of this study was to determine the most efficient cropping pattern that supports fungal growth, and biodegradation of crop residues. Three Pleurotus species- P. ostreatus (‘Grey Dover’), P. pulmonarius, and P. eryngii and two shiitake mushroom (Lentinula edodes) strains LE001 (ATCC #20546) and LE002 (ATCC #20635) were used in the study. Tomato residue was obtained from greenhouse grown tomatoes while straw was obtained from field-grown rice and wheat and stored air-dried in the greenhouse. Lettuce residue was obtained from growth chamber grown plants at Kennedy Space Center, FL. Cowpea, soybean, sweet potato, and basil inedible residues were provided by Dr. Cary Mitchell’s lab at Purdue University. The dry residues were milled and passed through a 2mm sieve to obtain fine particle size substrates.

A. Mixed cropping. Processed crop residues (wheat, rice, soybean, cowpea, tomato, lettuce, sweet potato) were weighed and 11.5g of each crop residue were combined in single Plantcon® tissue culture containers (Sigma, St. Louis, MO) to make 80.5g of residue per container. The residues were thoroughly mixed, moistened, and autoclaved. They were seeded with three species of Pleurotus (P. ostreatus [‘Grey Dover’ and ‘Pohu’], P. eryngii, P. pulmonarius) and four strains of L. edodes (LE2, LE3, LE001, LE002) and incubated in the dark at 22±2°C.. P. ostreatus strains ‘Pohu’ and ‘Grey Dover’ were the most prolific strains for mycelial growth and colonization in mixed residues. Mixing crop residue is advantageous in that degradation can be more effective using only a few efficient strains therefore eliminating the need to screen large numbers of strains to select specific ones that degrade only certain crop residues. Furthermore, mixing crop residues helps dilute the secondary metabolites of some crop residue such as sweet potato and basil, which may be inhibitory to fungal growth, while gaining additional C/N from the cereals/legumes residues for growth. Oyster mushroom strains were better than shiitake strains under mixed cropping patterns.

B. Co-Culture. Fungal co-culture was used to verify if there were synergistic colonization and degradation of crop residue, or antagonistic activity by two prolific species of Pleurotus. Pleurotus ostreatus ‘Grey Dover’ and P. pulmonarius were cultured on wheat straw amended with food waste at 0, 20, 40 60, 80 and 100% (v/v). Agar blocks containing the pure fungi were placed side by side at the center of the substrate. The diameter of the circle formed by outgrowing mycelia from both strains was used to determine growth, while recognizable basidiocarps were counted as fruit bodies.

Co-culturing P. ostreatus ‘Grey Dover’ and P. pulmonarius enhanced mycelial growth and colonization, and basidiocarp production, compared to composting with single species. However, the rate of degradation as measured by degradation efficiency index (DEI), depended on the concentration of organic nutrients in the compost and not the strains used. Food waste amendment did not improve degradation of wheat straw. Food waste concentration of 40% and above significantly reduced degradation capability of the oyster mushroom species. Co-cropping fungal species had a synergistic effect on fungal growth and fruiting. Use of efficient cropping methods may enhance fungal growth and fruiting, thus increase the rate of biodegradation of the substrates and efficiency of biomass recycling.

Nitrogen amendment enhances edible white-rot fungal growth and biodegradation of containerized inedible crop residues:

The objective of this study was to evaluate the contribution of nitrogen amendment to mycelial growth, fruiting and degradation efficiency of strains of white rot fungi. Furthermore, to increase fruiting and expedite fungal colonization and biodegradation of crop residues, wheat straw was amended with food waste as a source of nitrogen. Food waste was obtained from Dr. Jim Alleman’s laboratory at Purdue University, and kept at room temperature until use. Edible fungal strains of three species - P. ostreatus (‘Grey Dover’ ‘Blue Dolphin’), P. eryngii, and P. pulmonarius, chosen based on their ability to grow and rapidly colonize several substrates such as wheat, rice, etc. were cultured in YMVSA amended with food waste at 0, 20, 30, 40 and 50 % (v/v), to screen for species tolerating high concentrations of food waste. Two species that exhibited rapid growth in media containing 50% food waste were used further to study mycelial growth, fruiting and degradation of wheat straw amended with various food waste concentrations. Food waste slurry was diluted with distilled, deionized water to 6 concentrations [0, 20, 40, 60, 80 and 100% (v/v)] that were each added to milled wheat straw at 94ml per 60g wheat residue per container. Each treatment concentration was replicated five times. The mixed residue was steam sterilized at 121°C and 1.1kgm-2 for 40 min. Cultures were seeded with 3x3mm agar blocks of pure cultures of either Pleurotus ostreatus (Grey Dover), or Pleurotus pulmonarius under aseptic conditions. The inoculated cultures were placed at 22±2°C and 8 hours light for growth and fruiting.

Of the three oyster mushroom species P. ostreatus ‘Grey Dover’, P. pulmonarius, and P. eryngii, P. ostreatus and P. pulmonarius had similar mycelial growth rates which were significantly greater than that of P. eryngii. Growth of all species decreased with increased concentration of food waste supplemented in the YMVSA medium. This decrease in growth could be caused by the increase in pH of the amended media to about 7.4. The optimum pH range for Pleurotus is about 5.5-6.5. When cultured individually, Pleurotus species were most efficient in biodegradation when food waste was supplemented at 80% (v/v) in wheat straw. However, in the dual cultures, addition of food waste to wheat straw did not improve degradation efficiency.

Similarly, food waste amendment did not significantly improve mycelial growth in both strains compared to the control. It is thought that lignocellulosic wastes contain low amount of available N (Silva et al., 2005). Therefore, N addition may be necessary to increase fungal growth and degradation of the substrate. However, N addition is not always beneficial, for example, Buswell et al. (1995) found that N suppressed Mn-dependent peroxidase (MnP) production in L. edodes.

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Presentation	Mitchell, C. "Strategies for Minimizing the Energetic Penalties of Maintaining Human Presence at an Extended-Stay Mars Base" Miami University Biology Seminar, Oct. 12, Miami University, OH. Oct-2005
Presentation	Nyochembeng, L.M., C.A. Beyl and R.P. Pacumbaba "Edible fungal growth and fruiting on composted containerized inedible crop biomass" Poster presentation at ASHS annual meeting held in Las Vegas, NV, July 18-21, 2005. Jul-2005
Presentation	Nyochembeng, L.M. and C.A. Beyl. "Use of edible white rot fungi to enhance cellulose, hemicellulose and lignin degradation of crop residues" Seminar given to faculty and students of the College of Agricultural, Natural and Environmental Science at Tuskegee University, March 30-31, 2005. Mar-2005
Presentation	Nyochembeng, L.M. and C.A. Beyl. "Using edible fungi to break down inedible crop residues on long-term space missions" Oral presentation at the University and Industry Consortium (UIC) 2005 spring meeting held in Huntsville, AL. Apr-2005
Presentation	Nyochembeng, L.M. and C.A. Beyl "Growth Response of Edible Fungi on Processed Crop Biomass and Food Waste Amended Rice Straw" ALS/NSCORT Summer Research Symposium, Purdue University Aug-2004
Presentation	Nyochembeng, L.M., C.A. Beyl and R.P. Pacumbaba. "Enhancing edible white rot fungal degradation and recycling of solid wastes by incorporation of urea and STAR effluent" NASA ALS/NSCORT EAC meeting, Howard University Nov-2004